

WEB SCRAPING AND SOCIAL MEDIA SCRAPING

Scraping with Selenium

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Outline for today

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 - Selenium
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- 2 Common issues in webscraping
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 - CAPTCHAs
 - How to fight bans and CAPTCHAs

Motivation for crawling frameworks

Static and dynamic pages

- A **static webpage** is delivered to the user's web browser exactly as it is stored.
- A **server-side dynamic webpage** is constructed by a server processing server-side scripts.
- A **client-side dynamic webpage** is rendered using HTML scripting executed in the browser. JavaScript and other scripting languages determine the site's structure.
- Websites without scripts (or with minimal scripting) are referred to as static because they do not change after being loaded.
- **Truly static pages are quite rare nowadays**, mostly found in small business or personal websites.

Scraping dynamic pages

- When scripts generate page content, sending requests alone may be ineffective as some elements of the output may not be visible.
- Automating a browser is often a suitable solution, commonly achieved using **Selenium**.

What is Selenium?

- Simply put, Selenium automates browsers.
- More specifically, Selenium allows you to **access a browser** and use programming scripts to **simulate user actions**.
- Selenium enables interaction with different parts of a webpage by accessing its HTML structure, making it easy to **parse** content.
- One of its key features is the ability to simulate user behavior, such as clicking, typing, and uploading files.
- You can create a script that processes a webpage, mimics real user interactions, and collects data during the process.

Why use Selenium?

- Selenium is particularly useful for accessing webpages that **require user interaction**.
- Its applications include:
 - web scraping,
 - logging in and filling out forms,
 - handling dynamic webpages,
 - automated testing.
- For web scraping, Selenium is a powerful tool for interacting with pages that are difficult to access using other techniques.

Time efficiency of Selenium scrapers

- Since Selenium simulates user behavior, scraping with it **typically takes longer than using requests with BeautifulSoup**.
- **Delays** may be necessary to allow the browser to load pages fully.
- Frequent data saving or printing operations can slow down Selenium, though periodic saves are essential to prevent data loss in case of crashes.
- The challenge is to balance delays - **ensuring efficiency without making scraping excessively slow**.
- Testing a Selenium scraper on a **small sample before full deployment is highly recommended**.

Memory efficiency of Selenium scrapers

- The best practices for scraping static pages also apply here.
- The scraper should efficiently collect and store data.
- It is crucial to minimize the risk of unexpected crashes and memory overuse.

Common issues in webscraping

Why do issues arise even if scraping is legal?

- Even when legal, web scraping is often restricted by website administrators.
- Administrators **prevent site slowdowns** and **ensure accessibility** for human users.
- As a result, scrapers must operate in ways that **avoid detection**.
- Detection of a scraper/crawler usually results in a **temporary (rarely permanent) ban or CAPTCHA challenge**.

Bans

- A **ban** blocks a user from accessing a website.
- Bans can be temporary or permanent.
- Permanent bans are rare; more commonly, access is restricted for 24 hours.

CAPTCHAs

- **CAPTCHA** stands for Completely Automated Public Turing test to tell Computers and Humans Apart.
- It forces users to complete tasks that are difficult for bots, such as:
 - clicking a checkbox,
 - typing distorted text,
 - selecting images with specific objects.

CAPTCHAs

☐

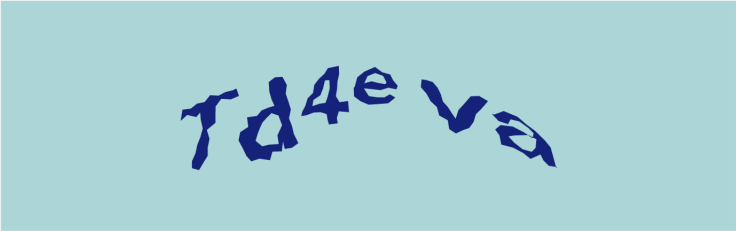
I'm not a robot

The reCAPTCHA logo consists of two interlocking arrows forming a circular shape. The top arrow is blue and points to the right, while the bottom arrow is grey and points to the left.

reCAPTCHA

[Privacy](#) - [Terms](#)

CAPTCHAs



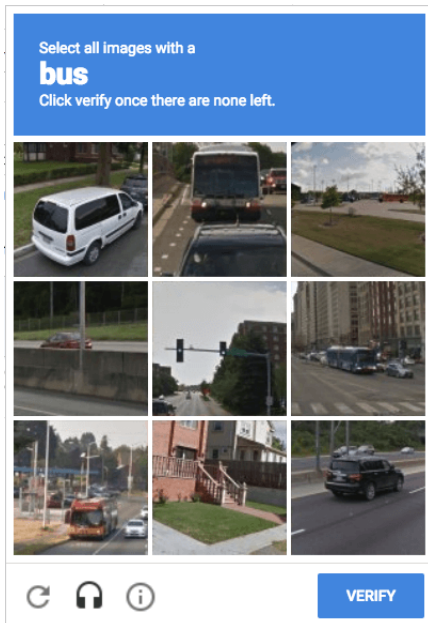
A CAPTCHA interface from Okta. It features a light blue rectangular box containing the distorted text "Td4e vā". Below this box, the text "Type the characters above:" is displayed. Underneath the text is a white input field. To the right of the input field is a dark blue button with the word "Go" in white. In the bottom left corner of the interface is the "okta" logo.

Type the characters above:

okta

Source: <https://www.okta.com/identity-101/captcha/>.

CAPTCHAs



Source: <https://onezero.medium.com/why-captcha-pictures-are-so-unbearably-depressing-20679b8cf84a>.

How to prevent bans and CAPTCHAs

- **Prevention is key** - avoiding detection is better than dealing with bans.
- **Understanding how scrapers are detected** is crucial.
- Users are typically identified by their **IP address, user-agent (device and browser ID), and behavioral patterns**.
- Administrators track interactions and attempt to distinguish bots from human users.

How are bans or CAPTCHAs triggered?

- Bots are often detected by **analyzing interaction patterns** - primitive bots click much faster than humans.
- Some websites use **trap links** - invisible to humans but embedded in the HTML structure. Bots that click everything will follow them and get flagged.

How to avoid bans and CAPTCHAs

- Avoid clicking every element on a webpage.
- **Introduce random delays** between actions, using a skewed distribution to mimic human behavior.
- **Rotate user-agents** (change browser and device IDs) to simulate different users.
- As a last resort, use **proxy servers** to rotate IP addresses.

What to do if you get banned or encounter CAPTCHA?

- If banned, try changing the device and browser ID first. If that fails, switch the IP address.
- Simple CAPTCHAs (e.g., clicking a checkbox) can sometimes be bypassed using automated scripts.
- Advanced CAPTCHAs requiring image recognition are difficult to bypass and often not worth the effort.

How to change IP address?

- Use a **proxy server**, which forwards requests while masking the real IP address.
- Rotating between multiple proxies helps avoid detection.
- Websites that block IPs based on request frequency can still detect and block all proxies in a pool—consider **premium proxy providers**.
- Keep in mind that bans may not be based on IP alone. Understanding the reason for detection is crucial.

Trivia: have you ever heard of companies offering manual CAPTCHA solving?

- This is a classic example of outsourcing. Spammers pay individuals with minimal IT skills to manually solve CAPTCHAs.
- This practice dates back more than 15 years, with costs typically ranging from \$0.75 to \$2.99 per 1,000 CAPTCHAs.
- Pricing depends on factors such as solving speed, service demand, and the number of available workers. Due to partial automation, some services now offer rates as low as \$0.40 per 1,000 CAPTCHAs.
- Depending on a website's policies, this practice **may be illegal**. Furthermore, the user—not the service provider—bears legal responsibility, as these companies often operate overseas and disregard user intent.

See: <https://www.nytimes.com/2010/04/26/technology/26captcha.html>,
<https://www.f5.com/labs/articles/cisotociso/i-was-a-human-captcha-solver>, <https://azcaptcha.com/pricing>.

Thank you for your attention!

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